



# Jesús Ignacio Castillo Barrios

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## Education

- 2023-2026 **Ph.D student in Science in Engineering and Computational Technologies**, CINVESTAV, Ciudad Victoria, Tamaulipas, *Mexico*.  
In progress
- 2020-2022 **Master of Science in Engineering and Computational Technologies**, CINVESTAV, Ciudad Victoria, Tamaulipas, *Mexico*.  
Certificate ID: 13843522
- 2015-2019 **Bachelor of Science in Computer Systems Engineering**, Instituto Tecnológico, Ciudad Valles, San Luis Potosí, *Mexico*.  
Certificate ID: 12019146

## Experience

- Jan 2024 – **Distributed Load Testing Engineer**, SIVEI (INTERNET VOTING SYSTEM).  
Jun 2024  
Performed distributed load testing and stress testing in a cluster using virtualization technology and Python scripting. Also utilized Selenium for automation.
- Details:
- o Implemented and managed load testing scenarios to evaluate system performance.
  - o Developed Python scripts for automating testing processes and analyzing results.
  - o Utilized Selenium for browser-based automation and functional testing.
  - o Analyzed and reported testing outcomes to improve system robustness and performance.
- Feb 2022 – **IT Audit Technician**, PREP SYSTEM - ELECTORAL INSTITUTE OF TAMAULIPAS.  
Jun 2022  
IT audit of the Operation Layer of the Preliminary Results Program (PREP) System 2022.

Jan 2021 – **Software Developer**, FORDECYT-PRONACES-CONACYT.

Feb 2022 Development of distributed systems and their deployment on private infrastructure for project 41756 “Technological platform for the management, assurance, exchange, and preservation of large volumes of health data and the construction of a national repository of health data analysis services.”

Details:

- Configuration and deployment of applications to production.
- Software product management.
- Generation of software documentation.

2019-2020 **Free lancer**, INDEPENDENT.

Software manager for several small-medium independent project. Focused on software management and software development

Detailed work:

- Scala backend development.
- Configuration and deploy of application in production.
- AWS administration.

2017-2019 **Software developer**, ONECLICK.

Software development for several independent projects. Focused on server side programming.

Detailed work:

- NodeJS backend development.
- Configuration and deploy of application in production.
- AWS administration.

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## Publications & Research

**Ignacio Castillo-Barrios, et al. (2026).** *APaC: A context-aware programming model and availability policy as code for elastic object storage.* [[10.1109/TCC.2026.3693716](https://doi.org/10.1109/TCC.2026.3693716)].

**Ignacio Castillo-Barrios, et al. (2025).** *MictlanX: Elastic code-defined object storage system* [[10.1145/3757347.3759148](https://doi.org/10.1145/3757347.3759148)].

**Ignacio Castillo-Barrios, et al. (2025).** *Code-and-Play with MictlanX, an elastic object storage framework for cloud services.* [Manuscript submitted for publication].

**Ignacio Castillo-Barrios, et al. (2025).** *CIATfunc: A policy-as-code and frame-based serverless model for composing blockchain-based and secure end-to-end applications.* [[10.1007/s10586-026-06108-7](https://doi.org/10.1007/s10586-026-06108-7)].

**Ignacio Castillo-Barrios, et al. (2024).** *Towards the Implementation of ONCA: A Generic, Scalable, and Massive Data Processing Platform for Information Discovery and Analytics.* [[10.1007/978-3-031-90203-1\\_16](https://doi.org/10.1007/978-3-031-90203-1_16)].

## Projects

**JUB Ecosystem**, *Software Architect*, <https://jub.tamps.cinvestav.mx/>.

Funded research project supported by SECIHTI under grant MADTEC-2025-M-478.

Contributions and technologies:

- o Vue for UI design and development.
- o Python FastAPI backend design and development.
- o MongoDB data modeling and custom query DSL implementation.
- o Asynchronous Python SDK client development.
- o Redis and Xolo authentication integration.

**Muyal-Ilal Platform**, *Software Architect*, <https://muyal.tamps.cinvestav.mx/>.

Contributions and technologies:

- o VUE 3 for UI Design / Development.
- o Python FAST API Backend Design / Development.
- o MongoDB Modeling Design / Implementation.
- o Docker Swarm Configuration.
- o Configuration and Deploy of Application in Production.

**ONCA**, *Software Architect*, <https://muyal.tamps.cinvestav.mx/observatories>.

Contributions and technologies:

- o VUE 3 for UI development.
- o Python FAST API for backend development.

**MictlanX - Object Storage System** , *Project Lead and Developer*, [GitHub Repository](#).

Contributions and technologies:

- o Python FAST API backend development.
- o Rust for core object storage system and file system manipulation.
- o Configuration and Deploy of Application in Production.

**Axo - Active Object System for Serverless Environments**, *Project Lead and Developer*, [GitHub Repository](#), [Demo](#) .

Contributions and technologies:

- o VUE 3 for UI development.
- o Python FAST API backend development.

## Languages

Spanish **Native language**

English **Intermediate**

*B1 level (523 points on TOEFL ITP)*